

Instructions:

You will work on these exercises during the practical sessions in the first week of the course.

Below is an indication of the exercises that we will cover (at least) in each session. We encourage you to try them before the session.

- Math4AI lecture (Tuesday): exercise 4.
- First tutorial session (Wednesday/Thursday): exercises 2(b,c), 3, and 7.
- Second tutorial session (Friday): exercise 9.

Conventions:

When it comes to taking derivatives with scalars, vectors, and matrices, we stick to the following conventions (these apply to all practice/homework sets):

- **Scalars, matrices and vectors:**
 - **Scalars** are denoted in the usual math font, e.g., $x \in \mathbb{R}$.
 - **Vectors** are denoted with boldface lower case, e.g., $\mathbf{x} \in \mathbb{R}^n$. If we want to be explicit about the objects being column vectors we may write $\mathbf{x} \in \mathbb{R}^{n \times 1}$ (with n rows and 1 column), and for row vectors we can say $\mathbf{x} \in \mathbb{R}^{1 \times n}$ (with 1 row and n columns). We rarely define a vector as a row vector but often write $\mathbf{x}^T$ to turn a column vector $\mathbf{x} \in \mathbb{R}^{n \times 1}$ into a row vector. The individual elements may be given by x_i , with i the row index.
 - **Matrices** are denoted with boldface upper case, e.g., $\mathbf{X} \in \mathbb{R}^{m \times n}$. Such a matrix has m rows and n columns. The elements of the matrices may be denoted with X_{ij} , with i the row index and j the column index.
- **Derivatives:** The derivative of a function $f(\mathbf{x})$ with respect to input $\mathbf{x}$ will be denoted using derivative notation. We then distinguish several cases for what kind of objects $\mathbf{x}$ and f are.
 - **Scalar input, scalar output:** $\frac{df(x)}{dx} \in \mathbb{R}$ when $x \in \mathbb{R}$ and $f(x) \in \mathbb{R}$.
 - **Vector input, scalar output:** $\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} \in \mathbb{R}^{1 \times n}$ when $\mathbf{x} \in \mathbb{R}^n$ and $f(\mathbf{x}) \in \mathbb{R}$. The result is thus a row vector.
 - **Vector input, vector output:** $\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} \in \mathbb{R}^{m \times n}$ when $\mathbf{x} \in \mathbb{R}^n$ and $f(\mathbf{x}) \in \mathbb{R}^m$. The result is a matrix in which each row represents the derivative $\frac{\partial f_i(\mathbf{x})}{\partial \mathbf{x}}$ of the i -th output of f . This matrix is known as the *Jacobian matrix*.
 - **Matrix input, scalar output:** $\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}} \in \mathbb{R}^{m \times n}$ when $\mathbf{X} \in \mathbb{R}^{m \times n}$ and $f(\mathbf{X}) \in \mathbb{R}$. The result is a matrix in which each element $\left[\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}} \right]_{ij} = \frac{\partial f}{\partial X_{ij}}(\mathbf{X})$ is the derivative of f with respect to matrix element X_{ij} .
- **Other conventions:** In other books and papers you may encounter different conventions. E.g., you may encounter the derivative of a scalar function to a vector input to be a column vector (instead of what we define to be a row vector). This is all fine as long as the conventions are consistently applied. We encourage you to stick to our recommended convention, but if you decide to deviate from it, please explicitly define your convention.

Exercise 1: Derivatives of scalar functions

Compute the derivatives of the following functions with respect to the input x :

- (a) $\sigma(x) = \frac{1}{1+e^{-x}}$ (the standard logistic function or "sigmoid function").

Solution:

To calculate the derivative of the *sigmoid function* we need to apply the chain rule:

$$\begin{aligned}\frac{d\sigma}{dx} &= (1 + e^{-x})^{-2} e^{-x} \\ &= \frac{e^{-x}}{(1 + e^{-x})^2} \\ &= \frac{e^{-x}}{(1 + e^{-x})} \sigma(x) \\ &= \left(\frac{e^{-x} + 1}{e^{-x} + 1} - \frac{1}{e^{-x} + 1} \right) \sigma(x) \\ &= \sigma(x)(1 - \sigma(x)).\end{aligned}$$

- (b) $\text{ReLU}(x) = \max\{0, x\}$ (this function is the "Rectified Linear Unit", a widely used activation function for neural networks).

Solution:

$$\frac{d\text{ReLU}}{dx} = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x < 0. \end{cases}$$

(note that the function is not differentiable at $x = 0$).

Exercise 2: Derivatives of multivariate functions

Calculate the following derivatives:

- (a) $\frac{df}{d\mathbf{x}}$, where $f(\mathbf{x}) = 2 \exp(x_2 - \log(x_1^{-1}) - \sin(x_3 x_1^2))$, $\mathbf{x} \in \mathbb{R}^3$.

Solution:

Given our convention the derivative will be a row vector, in particular:

$$\begin{aligned}\frac{d}{d\mathbf{x}} &= \begin{bmatrix} \frac{df(\mathbf{x})}{dx_1} \\ \frac{df(\mathbf{x})}{dx_2} \\ \frac{df(\mathbf{x})}{dx_3} \end{bmatrix}^T. \\ \frac{df(\mathbf{x})}{dx_1} &= f(\mathbf{x}) \left(\frac{1}{x_1} - 2x_3 x_1 \cos(x_3 x_1^2) \right). \\ \frac{df(\mathbf{x})}{dx_2} &= f(\mathbf{x}). \\ \frac{df(\mathbf{x})}{dx_3} &= -f(\mathbf{x}) \cos(x_3 x_1^2) x_1^2.\end{aligned}$$

- (b) $\frac{dh}{dy}$, where $h := h(\mathbf{x}) = x_1^3 + \exp(x_2)$ and $\mathbf{x} := \mathbf{x}(y) = [y \sin(y), y \cos(y)]^T$ and $y \in \mathbb{R}$. First show your understanding of the chain rule before plugging in the actual derivatives.

Solution:

By applying the chain rule we can write:

$$\begin{aligned} \frac{dh}{dy} &= \frac{dh}{d\mathbf{x}} \frac{d\mathbf{x}}{dy} \\ &= [3x_1^2 \quad \exp x_2] \begin{bmatrix} \sin(y) + y \cos(y) \\ \cos(y) - y \sin(y) \end{bmatrix} \\ &= [3y^2 \sin(y)^2 \quad \exp(y \cos(y))] \begin{bmatrix} \sin(y) + y \cos(y) \\ \cos(y) - y \sin(y) \end{bmatrix} \\ &= 3y^2 \sin(y)^2 (\sin(y) + y \cos(y)) + \exp(y \cos(y)) (\cos(y) - y \sin(y)). \end{aligned}$$

- (c) $\frac{dh}{d\mathbf{a}}$, where h is the same as in the previous subquestion, $\mathbf{a} := [y, z]^T \in \mathbb{R}^2$ and $\mathbf{x} := \mathbf{x}(\mathbf{a}) = [y \sin(y) + z, y \cos(y) + z^2]^T$.

Solution:

By applying the chain rule, we get:

$$\frac{dh}{d\mathbf{a}} = \frac{dh}{d\mathbf{x}} \frac{d\mathbf{x}}{d\mathbf{a}}.$$

The first part will be $\mathbb{R}^{1 \times 2}$ and the second will be $\mathbb{R}^{2 \times 2}$ resulting in a final shape of 1×2 .

$$\begin{aligned} \frac{dh}{d\mathbf{x}} &= [3x_1^2 \quad \exp(x_2)] \\ \frac{d\mathbf{x}}{d\mathbf{a}} &= \begin{bmatrix} \sin(y) + y \cos(y) & 1 \\ \cos(y) - y \sin(y) & 2z \end{bmatrix}. \end{aligned}$$

Exercise 3: Derivative shapes

What is the shape of the following derivatives according to the convention we follow in the course?

- (a) $\frac{df}{dx}$, where $f : \mathbb{R} \rightarrow \mathbb{R}$ and $x \in \mathbb{R}$.

Solution:

$$\frac{df}{dx} \in \mathbb{R}.$$

- (b) $\frac{df}{d\mathbf{x}}$, where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$.

Solution:

$$\frac{df}{d\mathbf{x}} \in \mathbb{R}^{1 \times n} \text{ (a row vector).}$$

- (c) $\nabla_{\mathbf{x}} f$, where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$.

Solution:

$$\nabla_{\mathbf{x}} f \in \mathbb{R}^n, \text{ or, in matrix form, } \frac{df}{d\mathbf{x}} \in \mathbb{R}^{n \times 1} \text{ (a column vector).}$$

- (d) $\frac{df}{dx}$, where $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$.

Solution:

$$\frac{df}{dx} \in \mathbb{R}^{m \times n}.$$

Exercise 4: Multivariate calculus

The following questions are good practice in manipulating vectors and matrices and they are essential for solving for posterior distributions. We consider the following expression:

$$(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{S}^{-1} (\mathbf{x} - \boldsymbol{\mu}).$$

where $\mathbf{x}$, $\boldsymbol{\mu}$, are vectors and $\mathbf{S}^{-1}$ is a symmetric, positive definite and therefore invertible matrix.

- (a) Expand the expression and gather terms.

Solution:

Firstly, it will be beneficial for us to identify the dimensionality of each of the terms in the expression. Hence, $\mathbf{x}, \boldsymbol{\mu} \in \mathbb{R}^n$ and $\mathbf{S} \in \mathbb{R}^{n \times n}$.

Then, due to the distributive property, we can expand the expression:

$$\mathbf{x}^T \mathbf{S}^{-1} \mathbf{x} - \mathbf{x}^T \mathbf{S}^{-1} \boldsymbol{\mu} - \boldsymbol{\mu}^T \mathbf{S}^{-1} \mathbf{x} + \boldsymbol{\mu}^T \mathbf{S}^{-1} \boldsymbol{\mu}.$$

Noticing that the transpose of a scalar is the scalar itself, we can combine the two middle terms.

$$\mathbf{x}^T \mathbf{S}^{-1} \mathbf{x} - 2\mathbf{x}^T \mathbf{S}^{-1} \boldsymbol{\mu} + \boldsymbol{\mu}^T \mathbf{S}^{-1} \boldsymbol{\mu}.$$

- (b) Compute the following derivatives, which you will use in the next part:

- (i) $\frac{d}{dx} \mathbf{v}^T \mathbf{x}$
(ii) $\frac{d}{dx} \mathbf{x}^T \mathbf{A} \mathbf{x}$.

Solution:

Let us first compute (i). We need to calculate the derivative of $f = \mathbf{v}^T \mathbf{x}$ using *index notation*. Firstly, we can write f as:

$$\sum_{k=1}^n v_k x_k.$$

Then, we need to calculate $\frac{\partial f}{\partial x_i}$, which is v_i . Finally, we can conclude that $\frac{d}{dx} f = \mathbf{v}^T$.

Let us now compute (ii). We need to calculate the derivative of $f = \mathbf{x}^T \mathbf{A} \mathbf{x} \in \mathbb{R}$ using *index notation*. Firstly, we can write f as:

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = \sum_{i=1}^n x_i \sum_{j=1}^n A_{ij} x_j.$$

Then we need to calculate $\frac{\partial f}{\partial x_k}$:

$$\frac{\partial f}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^n x_i \sum_{j=1}^n A_{ij} x_j.$$

Next step is to apply the product rule, and therefore we can write:

$$\frac{\partial f}{\partial x_k} = \sum_{i=1}^n \frac{\partial x_i}{\partial x_k} \sum_{j=1}^n A_{ij} x_j + \sum_{i=1}^n x_i \sum_{j=1}^n A_{ij} \frac{\partial x_j}{\partial x_k}.$$

Since $\frac{\partial x_i}{\partial x_k} = 1$ when $i = k$ and zero otherwise, for clarity, we can make use of Kronecker delta:

$$= \sum_{i=1}^n \delta_{ki} \sum_{j=1}^n A_{ij} x_j + \sum_{i=1}^n x_i \sum_{j=1}^n A_{ij} \delta_{kj}.$$

which gives the final expression for the derivative $\frac{\partial f}{\partial x_k}$:

$$= \sum_{j=1}^n A_{kj} x_j + \sum_{i=1}^n x_i A_{ik}.$$

Note that the first term is the dot product between vector $\mathbf{x}$ and the k -th row of matrix $\mathbf{A}$ while the second term is the dot product between vector $\mathbf{x}$ with the k -th column of the matrix $\mathbf{A}$. Now if we arrange all the partial derivatives in a row vector:

$$\begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \dots \\ \frac{\partial f}{\partial x_k} \\ \dots \\ \frac{\partial f}{\partial x_N} \end{bmatrix}^T = \begin{bmatrix} \sum_{j=1}^n x_j A_{1j} + \sum_{i=1}^n x_i A_{i1} \\ \sum_{j=1}^n x_j A_{2j} + \sum_{i=1}^n x_i A_{i2} \\ \dots \\ \sum_{j=1}^n x_j A_{kj} + \sum_{i=1}^n x_i A_{ik} \\ \dots \\ \sum_{j=1}^n x_j A_{Nj} + \sum_{i=1}^n x_i A_{iN} \end{bmatrix}^T = \mathbf{x}^T \mathbf{A}^T + \mathbf{x}^T \mathbf{A} = \mathbf{x}^T (\mathbf{A} + \mathbf{A}^T).$$

- (c) Use the two basic derivatives from the previous question to take the derivative with respect to $\boldsymbol{\mu}$, set it equal to 0, and solve for $\boldsymbol{\mu}$.

Solution:

$$f(\boldsymbol{\mu}) = \mathbf{x}^T \mathbf{S}^{-1} \mathbf{x} - 2\mathbf{x}^T \mathbf{S}^{-1} \boldsymbol{\mu} + \boldsymbol{\mu}^T \mathbf{S}^{-1} \boldsymbol{\mu}.$$

By applying (i) to the linear term in $\boldsymbol{\mu}$ and (ii) to the quadratic term in $\boldsymbol{\mu}$, noting that $\mathbf{S}^{-1}$ is symmetric, we get that

$$\frac{d}{d\boldsymbol{\mu}} f = -2\mathbf{x}^T \mathbf{S}^{-1} + 2\boldsymbol{\mu}^T \mathbf{S}^{-1}.$$

Setting it equal to zero, canceling the twos, and multiplying by $\mathbf{S}$ we get that the unique solution is $\boldsymbol{\mu} = \mathbf{x}$. Note that the last step is possible because $\mathbf{S}$ (and therefore, $\mathbf{S}^{-1}$) is invertible.

Note: Fun fact, we could have solved this exercise in an easier way, without expanding anything or taking a single derivative. We know that $\mathbf{S}^{-1}$ is PD, so $\mathbf{v}^T \mathbf{S}^{-1} \mathbf{v} > 0$ for all $\mathbf{x} \neq \mathbf{0}$. Since $\boldsymbol{\mu} = \mathbf{x}$ makes the expression zero, it has to be the minimum!

Exercise 5: Probability of events

Consider the following setting. You are driving down the street at night and suddenly you see a man climbing through a broken window of a jewelry store. Then, he runs away carrying a bag over his shoulder. For many of us, our gut reaction would be to think the man in question is a criminal. Why do we draw this conclusion instead of another scenario? Let's explore this using the methods of Probability Theory.

- (a) Explain in words: why would many people draw the conclusion that the man in question is a criminal? Try to think in terms of probability (1-2 sentences are sufficient).

Solution:

Any reasonable argument would work here.

- (b) Show, formally (using probability theory), that the probability of us believing the man is a criminal given our observation is based on our beliefs of making this observation when the man is a criminal and making the observation when the man is not a criminal. Define first your variables for the problem and then your answer.

Solution:

$\{C, O\}$. Let us define the event of the observations as O and the event of the man being a criminal as C . Then $P(C|O)$ is the probability of the man being a criminal given our observation. Using Bayes rule, we find:

$$P(C|O) = \frac{P(O|C)P(C)}{P(O|C)P(C) + P(O|\neg C)P(\neg C)}.$$

Clearly, our beliefs of making the observation given that the man is either a criminal or not a criminal play a role here.

- (c) Let's assume one in every 10^5 people is in fact a criminal, the probability of making this observation when the man is not a criminal is $\frac{1}{10^6}$, and that of making this observation when the man is a criminal is 0.8.

Compute the probability of the man being a criminal based on our observations.

Solution:

Using the numbers that were just given we have:

$$P(C|O) = \frac{P(O|C)P(C)}{P(O|C)P(C) + P(O|\neg C)P(\neg C)} = .. \approx 0.89.$$

- (d) The next morning you learn that a group of kids have smashed multiple store fronts in your neighborhood. How does this change your beliefs, i.e., do you still think the man is a criminal? Explicitly state which belief updates you make and re-compute the probability of the man being a criminal given our observation. *Note: you do not have to do a Bayesian update or justify your belief update mathematically.*

Solution:

Any argument and correct computation is considered correct. Since kids smashed the windows in store fronts, it is more likely to see a non-criminal in the situation we observed. Therefore, we could update our belief of an innocent man being in this situation. E.g.,

$P(O|\neg C) = 0.01$, which results in:

$$P(C|O) = \frac{P(O|C)P(C)}{P(O|C)P(C) + P(O|\neg C)P(\neg C)} = .. \approx 0.0008.$$

Exercise 6: Posterior distribution

For this question, you will compute the expression for the posterior parameter distribution for a simple data problem. Assume we observe N univariate data points $x_1, x_2, \dots, x_N$. Furthermore, we assume that they are generated by a Gaussian distribution with known variance σ^2 , but unknown mean μ . Assume a prior Gaussian distribution over the unknown mean, i.e., $p(\mu) = \mathcal{N}(\mu|\mu_0, \sigma_0^2)$. When answering these questions, use $\mathcal{N}(a|b, c^2)$ to indicate a Gaussian (normal) distribution over a with mean b and variance c^2 . You do not need to write down the explicit form of a Gaussian distribution.

- (a) Write down the general expression for a posterior distribution, using θ for the parameter, $\mathcal{D}$ for the data. Indicate the prior, likelihood, evidence, and posterior.

Solution:

$$\underbrace{p(\theta|D)}_{\text{posterior}} = \frac{\overbrace{p(\theta)}^{\text{prior}} \overbrace{p(D|\theta)}^{\text{likelihood}}}{\underbrace{p(D)}_{\text{evidence}}} = \frac{\overbrace{p(\theta)}^{\text{prior}} \overbrace{p(D|\theta)}^{\text{likelihood}}}{\underbrace{\int p(\theta)p(D|\theta)d\theta}_{\text{evidence}}}.$$

- (b) Write the posterior for this particular example. You do not need an analytic solution.

Solution:

$$p(\theta|D) = \frac{\mathcal{N}(\mu|\mu_0, \sigma_0^2) \prod_{n=1}^N \mathcal{N}(x_n|\mu, \sigma^2)}{\int \mathcal{N}(\mu|\mu_0, \sigma_0^2) \prod_{n=1}^N \mathcal{N}(x_n|\mu, \sigma^2) d\mu}.$$

Exercise 7: MLE of the Exponential distribution

We consider i.i.d. samples $x_1, \dots, x_n$ from an exponential distribution with rate parameter $\lambda > 0$:

$$f(x | \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

- (a) Derive the maximum likelihood estimator $\hat{\lambda}_{\text{ML}}$.

Note: In this exercise, prove that the critical point you obtain is indeed a maximum by checking the concavity of the log-likelihood.

Solution:

First, define the likelihood:

$$p(\mathbf{x} | \lambda) \stackrel{\text{i.i.d.}}{=} \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}.$$

The MLE is the parameter value that maximizes this likelihood:

$$\hat{\lambda}_{\text{ML}} = \arg \max_{\lambda} p(\mathbf{x} | \lambda) = \arg \max_{\lambda} \log p(\mathbf{x} | \lambda).$$

Taking logs gives:

$$\log p(\mathbf{x} | \lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i.$$

To find the optimum, differentiate the log-likelihood and set the derivative to zero:

$$\frac{d \log p(\mathbf{x} | \lambda)}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i \stackrel{!}{=} 0.$$

Hence,

$$\hat{\lambda}_{\text{ML}} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}.$$

Since

$$\frac{d^2 \log p(\mathbf{x} | \lambda)}{d\lambda^2} = -\frac{n}{\lambda^2} < 0,$$

the log-likelihood is concave and this critical point is indeed a maximum.

- (b) For the dataset $\mathbf{x} = (0.5, 1.2, 0.7, 1.6)$, compute $\hat{\lambda}_{\text{ML}}$.

Solution:

We have $n = 4$ and

$$\sum_{i=1}^4 x_i = 0.5 + 1.2 + 0.7 + 1.6 = 4.0.$$

Therefore,

$$\hat{\lambda}_{\text{ML}} = \frac{4}{4.0} = 1.0.$$

- (c) As an alternative to the analytical derivation we did in the previous subquestions, we can estimate the optimal value of λ using a numerical optimization algorithm. Apply gradient descent to find the global minimum of the negative log-likelihood. Assume a randomly initialized value of $\lambda_0 = 0.2723$ and use a learning rate of $\eta = 0.1$. Stop when the absolute change in log-likelihood between two consecutive iterations is smaller than 0.01. How many iterations are needed for convergence? How does the result compare to the analytical one that you obtained in question (b)?

Note: feel free to use a script or a spreadsheet to solve this exercise.

Solution:

For the sample above, $n = 4$ and $\sum_i x_i = 4$, so the negative log-likelihood is

$$J(\lambda) = -\log p(\mathbf{x} | \lambda) = -4 \log \lambda + 4\lambda.$$

Then

$$\frac{dJ}{d\lambda} = -\frac{4}{\lambda} + 4,$$

and the gradient-descent update is

$$\lambda_{t+1} = \lambda_t - \eta \frac{dJ}{d\lambda_t} = \lambda_t + 0.1 \left(\frac{4}{\lambda_t} - 4 \right).$$

Starting with $\lambda_0 = 0.2723$, we obtain:

Iteration (t)	λ_t	$J(\lambda_t)$	$\frac{\partial J}{\partial \lambda}(\lambda_t)$	Converged?
0	0.2723	6.2926	-10.6897	–
1	1.3413	4.1906	1.0177	FALSE
2	1.2395	4.0992	0.7729	FALSE
3	1.1622	4.0475	0.5583	FALSE
4	1.1064	4.0211	0.3846	FALSE
5	1.0679	4.0088	0.2544	FALSE
6	1.0425	4.0035	0.1630	TRUE

([Spreadsheet](#))

So the algorithm converges in 6 iterations, obtaining the estimate

$$\hat{\lambda}_{\text{ML}} = \lambda_9 = 1.0425,$$

which is close to the analytical MLE from (b) ($\hat{\lambda}_{\text{ML}} = 1.0$).

Exercise 8: MLE of the Rayleigh distribution

Exam question

A meteorologist is analyzing wind speed measurements from a coastal weather station. The wind speed X (in m/s) follows a Rayleigh distribution, which has the following probability density function:

$$p(x | \theta) = \frac{x}{\theta^2} \exp\left(-\frac{x^2}{2\theta^2}\right), \quad x \geq 0, \theta > 0.$$

The meteorologist gathers a set of independent wind speed measurements $\mathbf{x} = \{x_1, x_2, \dots, x_N\}$.

(a) Compute the log-likelihood of the data $\mathbf{x}$ under the parameter θ .

Solution:

The likelihood function is

$$p(\mathbf{x} | \theta) \stackrel{\text{i.i.d.}}{=} \prod_{i=1}^N \frac{x_i}{\theta^2} \exp\left(-\frac{x_i^2}{2\theta^2}\right).$$

Therefore the log-likelihood is

$$\log p(\mathbf{x} | \theta) = \sum_{i=1}^N \log x_i - 2N \log \theta - \frac{1}{2\theta^2} \sum_{i=1}^N x_i^2.$$

(b) Compute the maximum likelihood estimator for θ^2 .

Solution:

The MLE is the parameter value that maximizes the likelihood, or equivalently its logarithm:

$$\hat{\theta}_{\text{ML}}^2 = \arg \max_{\theta^2} p(\mathbf{x} \mid \theta) = \arg \max_{\theta^2} \log p(\mathbf{x} \mid \theta).$$

We then compute the derivative of the log-likelihood and set it to zero to find the optimal value.

Option 1 - optimizing w.r.t. θ :

Applying the chain rule, we obtain

$$\frac{d \log p(\mathbf{x} \mid \theta)}{d\theta} = \frac{d \log p(\mathbf{x} \mid \theta)}{d\theta^2} \cdot \frac{d\theta^2}{d\theta} = \frac{d \log p(\mathbf{x} \mid \theta)}{d\theta^2} \cdot 2\theta.$$

Since $\theta > 0$, the mapping $\theta \mapsto \theta^2$ is strictly increasing. Therefore, the location and nature of the critical point of $\log p(\mathbf{x} \mid \theta)$ remain unchanged no matter if we differentiate with respect to θ or θ^2 .

The derivative of the log-likelihood with respect to θ is

$$\frac{d \log p(\mathbf{x} \mid \theta)}{d\theta} = -\frac{2N}{\theta} + \frac{\sum_{i=1}^N x_i^2}{\theta^3}.$$

Setting the derivative to 0 and solving for θ^2 , we obtain

$$\frac{2N}{\theta} = \frac{\sum_{i=1}^N x_i^2}{\theta^3} \Rightarrow \hat{\theta}_{\text{ML}}^2 = \frac{1}{2N} \sum_{i=1}^N x_i^2.$$

Option 2 - reparametrizing:

Equivalently, let $s = \theta^2$. Rewriting the same log-likelihood in terms of s gives

$$\log p(\mathbf{x} \mid s) = \sum_{i=1}^N \log x_i - N \log s - \frac{1}{2s} \sum_{i=1}^N x_i^2.$$

The derivative of the log-likelihood with respect to s is

$$\frac{d \log p(\mathbf{x} \mid s)}{ds} = -\frac{N}{s} + \frac{1}{2s^2} \sum_{i=1}^N x_i^2.$$

Setting the derivative to 0 and solving for s , we obtain

$$\hat{s}_{\text{ML}} = \frac{1}{2N} \sum_{i=1}^N x_i^2.$$

(c) Show that if $x \sim \text{Rayleigh}(\theta)$, then $\mathbb{E}[x^2] = 2\theta^2$.

Hint: you can use the change of variable $t = \frac{x^2}{2\theta^2}$ to solve the integral.

Solution:

The second moment of X is

$$\mathbb{E}[X^2] = \int_0^\infty x^2 p(x \mid \theta) dx = \int_0^\infty \frac{x^3}{\theta^2} \exp\left(-\frac{x^2}{2\theta^2}\right) dx = \int_0^\infty x^2 \exp\left(-\frac{x^2}{2\theta^2}\right) \frac{x}{\theta^2} dx$$

Substituting $t = \frac{x^2}{2\theta^2}$, so that $dt = \frac{x}{\theta^2} dx$ and $x^2 = 2\theta^2 t$, we obtain

$$\mathbb{E}[x^2] = \int_0^\infty 2\theta^2 t e^{-t} dt = 2\theta^2 \int_0^\infty t e^{-t} dt.$$

We compute the remaining integral by parts. Let $u = t$ so that $du = dt$, and let $dv = e^{-t} dt$ so that $v = -e^{-t}$. Then

$$\int_0^\infty t e^{-t} dt = [-te^{-t}]_0^\infty + \int_0^\infty e^{-t} dt.$$

The boundary term vanishes since $\lim_{t \rightarrow \infty} te^{-t} = 0$, and at $t = 0$ it is also 0. Hence

$$\int_0^\infty t e^{-t} dt = \int_0^\infty e^{-t} dt = [-e^{-t}]_0^\infty = 1.$$

- (d) Prove that the estimator that you have computed in question (b) is unbiased. *Hint: use the result from the previous subquestion.*

Solution:

We want to show that the maximum likelihood estimator

$$\hat{\theta}_{\text{ML}}^2 = \frac{1}{2N} \sum_{i=1}^N x_i^2$$

is unbiased, i.e. $\mathbb{E}[\hat{\theta}_{\text{ML}}^2] = \theta^2$.

By linearity of expectation,

$$\mathbb{E}[\hat{\theta}_{\text{ML}}^2] = \mathbb{E}\left(\frac{1}{2N} \sum_{i=1}^N x_i^2\right) = \frac{1}{2N} \sum_{i=1}^N \mathbb{E}[x_i^2] = \frac{1}{2} \mathbb{E}[x^2],$$

where $X \sim \text{Rayleigh}(\theta)$.

Using the hint, we obtain:

$$\mathbb{E}[\hat{\theta}_{\text{ML}}^2] = \frac{1}{2} \mathbb{E}[x^2] = \theta^2.$$

Thus, the estimator $\hat{\theta}_{\text{ML}}^2$ is unbiased.

Exercise 9: MLE and MAP of the Binomial distribution

Suppose someone gave you a magic coin that even though it is perfectly flat, it shows heads much more than tails. You toss it 8 times, resulting in an 8-bit binary string.

- (a) This sequence can be modelled with a binomial distribution. If the coin is unbiased, what will be the value of p ?

Solution:

If the coin is truly unbiased it will have value of exactly 0.5 and thus we would expect 4 1s and 4 0s. Thus, for an unbiased coin, $p = 0.5$.

- (b) We observe the bitstring 01110110. What is the maximum likelihood estimate of p ? First derive the question analytically, then give the numerical solution.

Solution:

First, define the likelihood. Here, f denotes the probability mass function, while p denotes the coin parameter.

$$f(m | n, p) = \binom{n}{m} p^m (1-p)^{n-m}.$$

The MLE is the parameter value that maximizes this likelihood:

$$\hat{p}_{\text{ML}} = \arg \max_p f(m | n, p) = \arg \max_p \log f(m | n, p).$$

We compute the derivative of the log-likelihood and set it to zero to find the optimum.

$$\frac{d}{dp} \log f(m | n, p) = \frac{m}{p} - \frac{n-m}{1-p} \stackrel{!}{=} 0.$$

$$\Leftrightarrow m(1-p) = p(n-m).$$

$$\Leftrightarrow m = pn.$$

$$\Leftrightarrow \hat{p}_{\text{ML}} = \frac{m}{n}.$$

Insert numbers.

$$\hat{p}_{\text{ML}} = 5/8 = 0.625.$$

- (c) Knowing this, a naive person would accept the coin as biased. You are not naive, and you've flipped enough coins to know that almost all of them are practically unbiased. You express this by having a strong prior. This prior appears in Bayes' rule. It's a beta distribution, which is the conjugate prior of the binomial, resulting in a beta posterior distribution.

Show that the posterior distribution is beta.

Solution:

By Bayes' rule, the posterior distribution is proportional to the likelihood times the prior (the evidence term in the denominator does not depend on p):

$$f(p | m, n, a, b) \propto f(m | n, p) f(p | a, b)$$

The likelihood (Binomial) as a function of p is proportional to:

$$f(m | n, p) \propto p^m (1-p)^{n-m}$$

And the prior (Beta) is proportional to:

$$f(p | a, b) \propto p^{a-1} (1-p)^{b-1}$$

Multiplying these together, we find form of the posterior up to its normalization constant:

$$\begin{aligned} f(p | m, n, a, b) &\propto (p^m (1-p)^{n-m}) (p^{a-1} (1-p)^{b-1}) \\ &\propto p^{m+a-1} (1-p)^{n-m+b-1} \end{aligned}$$

We recognize this final expression as a Beta distribution with updated parameters $a' = m + a$ and $b' = n - m + b$.

Because any valid probability distribution must integrate to 1, there is only one possible normalizing constant that can turn this expression into a valid distribution. Thus, without needing to calculate the evidence explicitly, we know the posterior must be exactly that Beta distribution:

$$f(p \mid m, n, a, b) = \text{Beta}(p \mid a', b')$$

- (d) Incorporating your strong beta prior with $a = 13$ and $b = 13$, what is your skeptical (MAP) estimate of p ? Again, first derive analytically, then plug in the numbers.

Solution:

First, define the posterior objective obtained by multiplying the likelihood and prior:

$$f(p \mid m, n, a, b) \propto f(m \mid n, p) f(p \mid a, b) \propto p^{m+a-1} (1-p)^{n-m+b-1}.$$

The MAP estimate maximizes this posterior, or equivalently its logarithm:

$$\hat{p}_{\text{MAP}} = \arg \max_p f(p \mid m, n, a, b) = \arg \max_p \log f(p \mid m, n, a, b).$$

We compute the derivative of the log-posterior and set it to zero to find the optimum:

$$\frac{d}{dp} \log f(p \mid m, n, a, b) = \frac{m+a-1}{p} - \frac{n-m+b-1}{(1-p)} = 0.$$

$$\Leftrightarrow (1-p)(m+a-1) = p(n-m+b-1).$$

$$\Leftrightarrow \hat{p}_{\text{MAP}} = \frac{m+a-1}{n+b+a-2}.$$

Plug in the numbers.

$$\hat{p}_{\text{MAP}} = \frac{5+13-1}{8+13+13-2} = 0.53125.$$

- (e) How can a and b be interpreted?

After better inspection, you see that the coin has been made using two different kinds of metal. It seems clear now that this is not a regular coin. How does your belief change? What parameters change, and in which direction?

Solution:

a and b can be interpreted as extra datapoints (where a denotes the successes and $a+b$ the total observations) that you previously observed. Any other sensible interpretation should also be fine.

After the inspection, your prior becomes less strong, as you have not seen such a coin before, therefore a and b both become smaller, making the distribution less sharply peaked.

- (f) Could these results have been derived with a Bernoulli distribution, too? If so, explain why and obtain the ML (maximum likelihood) estimate.

Solution:

Since we independently conducted 8 trials, and we are only interested in obtaining the parameter p (we saw that when differentiating the binomial coefficient immediately dropped), this could have been modelled with 8 factorised Bernoulli distributions. First, define the likelihood:

$$f(D | p) \stackrel{\text{i.i.d.}}{=} \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}.$$

The MLE maximizes this likelihood, or equivalently its logarithm:

$$\hat{p}_{\text{ML}} = \arg \max_p f(D | p) = \arg \max_p \log f(D | p).$$

We compute the derivative of the log-likelihood and set it to zero to find the optimum:

$$\frac{d}{dp} \log f(D | p) = \sum_{j=1}^n xp - (1-x)(1-p) = 0.$$

$$\Leftrightarrow \hat{p}_{\text{ML}} = \frac{1}{n} \sum_{j=1}^n x_j = \frac{m}{n}.$$

And so we obtain the same result.

Exercise 10: Probability Distributions, Likelihoods, and Estimators

You live in Den Helder and find that it rains quite a lot. Your goal is to estimate the probability that it will rain on any given day of the year. For one year, for each month, you count the number of days with rain. You get the following counts from January to December:

21, 18, 17, 15, 13, 12, 15, 15, 18, 20, 21, 22.

This gives a grand total of 207 days with rain. Let r_t be a binary random variable denoting the observation for day t in that year: $r_t = 1$ means it rained on day t , and $r_t = 0$ means it did not rain. We want to estimate the probability ρ of rain on any day of the year. The number of rainy days per month is not important for the following questions; only the total for the year is relevant.

- (a) What is the likelihood for a single observation r_t ? What is the likelihood for the entire set of observations $\{r_t\}_{t=1}^N$? Use n_1 for the number of rainy days, n_0 for the number of days without rain, and N for the total number of days.

Solution:

For a single Bernoulli observation,

$$p(r_t | \rho) = \rho^{r_t} (1-\rho)^{1-r_t}.$$

For the full dataset,

$$p(\mathbf{r} | \rho) = \prod_{t=1}^N \rho^{r_t} (1-\rho)^{1-r_t} = \rho^{\sum_{t=1}^N r_t} (1-\rho)^{\sum_{t=1}^N (1-r_t)} = \rho^{n_1} (1-\rho)^{n_0}.$$

- (b) Write the log-likelihood for the entire set of observations.

Solution:

Taking logarithms,

$$\log p(\mathbf{r} \mid \rho) = n_1 \log \rho + n_0 \log(1 - \rho).$$

- (c) Solve for the maximum likelihood estimate of ρ . Do it in general, using n_0 and n_1 , and for this specific case by plugging in the numbers.

Solution:

First, define the likelihood:

$$p(\mathbf{r} \mid \rho) = \rho^{n_1} (1 - \rho)^{n_0}.$$

The MLE maximizes this likelihood, or equivalently its logarithm:

$$\hat{\rho}_{\text{ML}} = \arg \max_{\rho} p(\mathbf{r} \mid \rho) = \arg \max_{\rho} \log p(\mathbf{r} \mid \rho).$$

Let

$$f(\rho) = \log p(\mathbf{r} \mid \rho) = n_1 \log \rho + n_0 \log(1 - \rho).$$

We compute the derivative of the log-likelihood and set it to zero to find the optimum:

$$\frac{\partial f}{\partial \rho} = \frac{n_1}{\rho} - \frac{n_0}{1 - \rho} \stackrel{!}{=} 0.$$

Thus

$$n_1(1 - \rho) = n_0 \rho \implies \rho_{\text{ML}} = \frac{n_1}{n_0 + n_1} = \frac{n_1}{N}.$$

In this case, $n_1 = 207$ and $N = 365$, so

$$\hat{\rho}_{\text{ML}} = \frac{207}{365}.$$

- (d) Assume a Beta prior for ρ with parameters a and b . Solve for the MAP estimate of ρ .

Solution:

First, define the likelihood and prior:

$$p(\mathbf{r} \mid \rho) = \rho^{n_1} (1 - \rho)^{n_0}, \quad p(\rho) \propto \rho^{a-1} (1 - \rho)^{b-1}.$$

The MAP estimate maximizes the posterior:

$$\hat{\rho}_{\text{MAP}} = \arg \max_{\rho} p(\rho \mid \mathbf{r}) = \arg \max_{\rho} p(\mathbf{r} \mid \rho) p(\rho).$$

Hence the log-posterior, up to constants independent of ρ , is

$$f(\rho) = (n_1 + a - 1) \log \rho + (n_0 + b - 1) \log(1 - \rho) + C.$$

We compute the derivative of the log-posterior and set it to zero to find the optimum:

$$\frac{n_1 + a - 1}{\rho} - \frac{n_0 + b - 1}{1 - \rho} \stackrel{!}{=} 0.$$

Therefore,

$$\hat{\rho}_{\text{MAP}} = \frac{n_1 + a - 1}{N + a + b - 2}.$$

- (e) Write the form of the posterior distribution for ρ . You do not need to solve it analytically.

Solution:

By Bayes' rule,

$$p(\rho | \mathbf{r}) = \frac{p(\mathbf{r} | \rho)p(\rho)}{p(\mathbf{r})} = \frac{p(\mathbf{r} | \rho)p(\rho)}{\int_0^1 p(\mathbf{r} | \rho)p(\rho) d\rho}.$$

Since the likelihood is Bernoulli and the prior is Beta, the posterior is also Beta:

$$p(\rho | \mathbf{r}) = \text{Beta}(\rho | a + n_1, b + n_0).$$

- (f) Solve for the posterior distribution analytically. *Hint: it is a Beta distribution.*

Solution:

The likelihood and prior are proportional to

$$p(\mathbf{r} | \rho) \propto \rho^{n_1} (1 - \rho)^{n_0}, \quad p(\rho) \propto \rho^{a-1} (1 - \rho)^{b-1}.$$

Thus

$$p(\rho | \mathbf{r}) \propto \rho^{n_1+a-1} (1 - \rho)^{n_0+b-1}.$$

Normalizing gives

$$p(\rho | \mathbf{r}) = \frac{\Gamma(a+b+N)}{\Gamma(a+n_1)\Gamma(b+n_0)} \rho^{a+n_1-1} (1 - \rho)^{b+n_0-1},$$

which is $\text{Beta}(a + n_1, b + n_0)$.

Exercise 11: Maximum Likelihood Estimate of Angle Measurements

Find the maximum likelihood estimate of the angle θ , given two independent noisy measurements c and s , where c is a measure of the cosine and s is a measure of the sine of θ . Assume each measurement has known Gaussian standard deviation σ , the same in both cases. To make the solution unique, assume $\theta \in [-\pi/2, \pi/2]$.

Hint: Use $\sin^2 \theta + \cos^2 \theta = 1$ and gather all terms independent of θ into a constant C .

- (a) Write down the likelihood $p(s, c | \theta)$.

Solution:

By independence,

$$p(s, c | \theta) = p(s | \theta)p(c | \theta).$$

Since $s \sim \mathcal{N}(\sin \theta, \sigma^2)$ and $c \sim \mathcal{N}(\cos \theta, \sigma^2)$,

$$p(s, c | \theta) \propto \exp \left[-\frac{1}{2\sigma^2} (s - \sin \theta)^2 \right] \exp \left[-\frac{1}{2\sigma^2} (c - \cos \theta)^2 \right].$$

Thus

$$\begin{aligned} p(s, c \mid \theta) &\propto \exp \left[-\frac{1}{2\sigma^2} ((s - \sin \theta)^2 + (c - \cos \theta)^2) \right] \\ &= \exp \left[-\frac{1}{2\sigma^2} (s^2 + c^2 + 1 - 2s \sin \theta - 2c \cos \theta) \right] \\ &\propto \exp \left[\frac{1}{\sigma^2} (s \sin \theta + c \cos \theta) \right], \end{aligned}$$

where terms independent of θ are absorbed into the proportionality constant.

(b) Write down the log-likelihood.

Solution:

Up to constants independent of θ ,

$$\log p(s, c \mid \theta) = \frac{1}{\sigma^2} (s \sin \theta + c \cos \theta) + C.$$

(c) Obtain the maximum likelihood estimate.

Solution:

Using the likelihood from part (a), the MLE maximizes the likelihood, or equivalently its logarithm:

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta} p(s, c \mid \theta) = \arg \max_{\theta} \log p(s, c \mid \theta).$$

We compute the derivative of the log-likelihood with respect to θ and set it to zero to find the optimum:

$$\frac{\partial}{\partial \theta} \log p(s, c \mid \theta) = \frac{1}{\sigma^2} (s \cos \theta - c \sin \theta) \stackrel{!}{=} 0.$$

Hence

$$s \cos \theta = c \sin \theta.$$

If $c \neq 0$, this gives

$$\frac{s}{c} = \tan \theta.$$

Therefore, on the interval $[-\pi/2, \pi/2]$,

$$\hat{\theta}_{\text{ML}} = \arctan \left(\frac{s}{c} \right).$$